—in which the science stimulates the fiction, and the fiction stimulates a new generation of scientists, a process benefiting both genres. But as the next step in the dance, it is now clear that seeding Venus with special photosynthetic microorganisms will not work. Since 1961 we've discovered that the clouds of Venus are a concentrated solution of sulfuric acid, which makes the genetic engineering rather more challenging. But that in itself is not a fatal flaw. (There are microorganisms that live out their lives in concentrated solutions of sulfuric acid.) Here's the fatal flaw: In 1961 I thought the atmospheric pressure at the surface of Venus was a few "bars," a few times the surface pressure on Earth. We now know it to be 90 bars, so that if the scheme worked, the result would be a surface buried in hundreds of meters of fine graphite, and an atmosphere made of 65 bars of almost pure molecular oxygen. Whether we would first implode under the atmospheric pressure or spontaneously burst into flames in all that oxygen is an open question. However, long before so much oxygen could build up, the graphite would spontaneously burn back into $CO_2$, short-circuiting the process. At best, such a scheme can carry the terraforming of Venus only partway.

Let's assume that by the early twenty-second century we have comparatively inexpensive heavy-lift vehicles, so we can carry large payloads to other worlds; abundant and powerful fusion reactors; and well-developed genetic engineering. All three assumptions are likely, given current trends. Could we terraform the planets?* James Pollack of NASA's Ames Research Center and I surveyed this problem. Here's a summary of what we found:

VENUS: Clearly the problem with Venus is its massive greenhouse effect. If we could reduce the greenhouse effect almost to zero, the climate might be balmy. But a 90-bar $CO_2$ atmosphere is oppressively thick. Over every postage stamp-sized square inch of surface, the air weighs as much as six professional football players,